Special Optimal Thermal Cell: Correlator-Designed Supercells for Effective-Harmonic Phonons and Thermodynamics

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We introduce the Special Optimal Thermal Cell (SOTC), a deterministic displacement-design strategy for extracting effective second-order interatomic force constants (IFCs) from small periodic supercells. Inspired by the Special Quasirandom Structure construction, SOTC chooses collective atomic displacements by minimizing the mismatch between selected shell-averaged displacement correlators of the cell and those of a harmonic trial model. A self-consistent cycle updates the trial phonons from density-functional-theory (DFT) forces and ridge-regression IFC fits. For the nine systems studied here (FCC Al, Ag, Au, and Cu; BCC Mo and W; and rocksalt NaCl, MgO, and PbTe), the reported self-consistency criterion is reached in 2–5 iterations. The archived calculations use 1–8 displaced configurations per iteration and an undisplaced reference, corresponding to 4–36 force evaluations rather than one force evaluation per iteration. The resulting effective-harmonic phonons give constant-volume heat capacities within approximately 5% of experiment for the ionic compounds at their design temperatures and within 7% for all nine materials at 300 K. At the design temperatures of the metals, the harmonic heat capacity cannot reproduce electronic and explicit anharmonic contributions above the Dulong–Petit limit. The primary SOTC benchmarks assess second-order effective-harmonic and quasi-harmonic observables; they do not establish equivalence to methods that sample explicit anharmonic free-energy surfaces. In a separate 100-snapshot canonical validation for Al, NaCl, and W, symmetry-constrained FC2+FC3+FC4 models reduce held-out force RMSE by 72–94% relative to matched central finite-displacement phonopy FC2 baselines. This validates the higher-order fitting and export implementation, not a few-configuration SOTC sampling advantage. For NaCl, QE DFPT Born charges and the electronic dielectric tensor produce a 2.895 THz LO–TO splitting when applied to the matched finite-displacement phonons. The implementation is available as an open-source Python package interfaced to Quantum ESPRESSO.

I. INTRODUCTION

High-temperature phonon properties (renormalized dispersions, soft-mode instabilities, thermal expansion, heat capacity) encode the anharmonic response of a crystal to atomic vibrations with amplitudes reaching several percent of the lattice constant. Accurate first-principles descriptions of these properties are essential for predicting phase stability [1], thermal conductivity [2], electron–phonon coupling [3], and thermodynamic databases for materials design [4].

Finite-temperature lattice dynamics can be formulated at several distinct levels. The temperature-dependent effective potential framework fits effective second- and higher-order IFCs to finite-temperature force data [5–8]. Variational self-consistent harmonic methods optimize a trial distribution and its free energy [9–11], while mode-coupling theory provides a separate non-perturbative description of anharmonic lattice dynamics [12]. Ab initio molecular dynamics samples classical nuclear trajectories on the chosen electronic potential-energy surface [13]. These approaches do not target identical observables or approximations; meaningful cost or accuracy comparisons therefore require matched cells, electronic settings, temperatures, and physical outputs.

In a parallel development, Zunger and coworkers [14] demonstrated that a random substitutional alloy can be mimicked statistically by a single small supercell, the Special Quasirandom Structure (SQS), whose chemical correlation functions match those of the ideal random alloy up to a spatial cutoff. The key observation is that structure encodes information: the correct spatial correlations are imposed by construction rather than recovered by averaging over an ensemble.

We apply the special-structure idea to displacement design. A harmonic trial model defines displacement–displacement correlations $C_{ab}^{\alpha\beta}(\mathbf{R}, T)$. SOTC searches a fixed periodic supercell for a collective displacement pattern whose selected shell-averaged correlators approximate these targets up to the IFC cutoff. DFT forces on the designed configuration are then used in a symmetry-constrained ridge fit of second-order IFCs, and the fitted phonons update the target correlators. The small iteration counts reported below are empirical observations for the present benchmarks, not a proof that a single configuration identifies an arbitrary IFC tensor.

SOTC is thus a deterministic configuration-design and effective-FC2 fitting procedure. It does not minimize an anharmonic variational free energy or generate a time trajectory. Its proposed distinction is the explicit optimization of prescribed spatial correlator moments before the force calculation, rather than a general replacement for ensemble or trajectory methods.

This paper is organized as follows. Section II presents the theoretical framework: thermal displacement correlators, the SOTC objective function, IFC extraction by

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ridge regression, the self-consistency loop, and the quasi-harmonic extension for continuous thermodynamic properties. Section III describes computational details. Section IV reports benchmarks for all nine materials, including thermal expansion for aluminum (Sec. IV.F). Section V discusses the method's range of validity, limitations, computational cost, and comparison with the phonopy quasi-harmonic approximation (QHA). Section VI concludes.

II. THEORY

A. Thermal Displacement Correlators

Consider a crystal with n_b basis atoms and $N_c = N_1N_2N_3$ primitive cells, so that the supercell contains n_bN_c atoms. Let $u_a^\alpha(\mathbf{R})$ be the displacement of basis atom a in cell $\mathbf{R}$. With the Fourier convention defined in Appendix A, the equal-time canonical correlator of a harmonic lattice is [15]:

$$C_{ab}^{\alpha\beta}(\mathbf{R},T) = \langle u_a^\alpha(\mathbf{0})\, u_b^\beta(\mathbf{R})\rangle_T = \frac{\hbar}{2N_c\sqrt{M_aM_b}} \sum_{\mathbf{q},s} \frac{e_a^{\alpha*}(\mathbf{q}s)\, e_b^\beta(\mathbf{q}s)}{\omega_{\mathbf{q}s}} \coth\!\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right) e^{-i\mathbf{q}\cdot\mathbf{R}}. \tag{1}$$

Here $\mathbf{q}$ runs over the N_c wave vectors commensurate with the Born–von Kármán cell, s labels a phonon branch, and the polarization vectors obey the normalization stated in Appendix A. The remaining symbols have their usual meanings. Equation (1) is standard harmonic-lattice theory; a derivation and the phase convention are given in Appendix A.

The coth factor encodes the full quantum Bose–Einstein statistics, smoothly interpolating between the zero-point limit ($T \to 0$: $\coth \to 1$) and the classical equipartition limit ($k_BT \gg \hbar\omega$: $\coth \to 2k_BT/\hbar\omega$).

To reduce the tensor correlator to the scalar target used in the present implementation, let $\mathcal{S}_R^{ab}$ denote the set of lattice vectors belonging to the crystallographic shell of radius R for basis pair (a,b), with multiplicity N_R^{ab}. The shell and Cartesian trace average is defined by

$$\bar{C}_2^{ab}(R,T) = \frac{1}{3N_R^{ab}} \sum_{\mathbf{R}'\in\mathcal{S}_R^{ab}} \sum_\alpha C_{ab}^{\alpha\alpha}(\mathbf{R}',T). \tag{2}$$

Thus R labels a discrete neighbor shell; it is not a continuum spherical average. Numerically equivalent distances are grouped using the distance tolerance reported with the calculation inputs. Species/basis-pair channels are retained separately. This scalar reduction discards anisotropic and off-diagonal correlation information, a limitation discussed in Sec. V.D.

For a monatomic cubic crystal, the on-site value is the one-component isotropic displacement parameter $U_{\rm iso} = \frac{1}{3}\langle|\mathbf{u}|^2\rangle$:

$$U_{\rm iso}(T) = \bar{C}_2(0,T) = \frac{\hbar}{6N_cM} \sum_{\mathbf{q},s} \frac{1}{\omega_{\mathbf{q}s}} \coth\!\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right). \tag{3}$$

Equation (3) follows from the trace of Eq. (1) and the polarization-vector completeness relation. The total three-dimensional MSD is $3U_{\rm iso}$, while crystallographic Debye–Waller factors use $B = 8\pi^2U_{\rm iso}$. For compounds, species-resolved displacements are evaluated from the full eigenvector-weighted mode sum rather than a species-averaged mass and the total DOS. The displacement norm used to design a SOTC configuration is documented separately from this one-component quantity to avoid a factor-of-three ambiguity.

B. The SOTC Objective Function

Let Ω be a candidate supercell of n atoms (with $n \ll N$) and let $\{\mathbf{u}_i\}$ be a set of atomic displacements assigned to those atoms. The **SOTC objective function** measures the mismatch between the selected correlators of a finite displacement vector and the harmonic trial targets:

$$\mathcal{Q}(\mathbf{u}) = \sum_{a,b} \sum_{R<r_{\rm cut}} w_{ab}(R) \left[\bar{C}_{2,ab}^{\rm cell}(R;\mathbf{u}) - \bar{C}_{2,ab}(R,T)\right]^2. \tag{4}$$

Here $\bar{C}_{2,ab}^{\rm cell}$ is formed with the same shell, trace, and species averaging as Eq. (2), and $r_{\rm cut}$ is the real-space IFC cutoff. Equation (4) is introduced in this work; its gradient is given in Appendix C. A positive distance weight may be chosen, for example

$$w_{ab}(R;\xi_w) = \exp(-R/\xi_w), \tag{5}$$

where ξ_w is a numerical weighting length whose adequacy must be tested by cutoff and dispersion convergence. We do not identify ξ_w with a universal phonon mean free path: a finite dynamical phonon lifetime does not, by itself, prove exponential decay of an equal-time displacement correlator. No lifetime-based exponential decay or universal T^{-1}/T^{-3} scaling law is assumed in this formulation.

Minimization of $\mathcal{Q}$ over the displacement field $\{\mathbf{u}_i\}$ subject to $|\mathbf{u}_i| \le u_{\rm rms}(T)$ is performed by gradient descent with random restarts. The initial guess is the harmonic displacement pattern obtained by sampling each phonon mode with amplitude $A_{\mathbf{q}s}^{(0)} = \sqrt{\hbar\coth(\hbar\omega_{\mathbf{q}s}/2k_BT)/2\omega_{\mathbf{q}s}}$. The optimization adjusts mode amplitudes and phase relations to minimize the correlator mismatch while keeping displacements physically bounded.

C. IFC Extraction by Ridge Regression

Let $F_i^\alpha(\mathbf{u})$ and $F_i^\alpha(\mathbf{0})$ be forces on the displaced and reference structures at the same cell. The response force $\Delta F_i^\alpha = F_i^\alpha(\mathbf{u}) - F_i^\alpha(\mathbf{0})$ removes any residual reference force. To harmonic order, the collective-displacement relation with translational invariance written explicitly is

$$\Delta F_i^\alpha = -\sum_{j,\beta} \Phi_{ij}^{\alpha\beta}\,(u_j^\beta - u_i^\beta) + \mathcal{O}(u^2). \tag{6}$$

This constitutes an underdetermined system when the number of IFC parameters N_{IFC} exceeds $3n$ (the dimension of the force vector). We use Tikhonov (ridge) regression to stabilize the inversion:

$$\hat{\mathbf{\Phi}} = -\left(\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I}\right)^{-1}\mathbf{G}^T\Delta\mathbf{F}. \tag{7}$$

The geometry matrix $\mathbf{G}$ contains the relative displacements in Eq. (6), $\Delta\mathbf{F}$ stacks the response forces, and λ is the ridge parameter. Equation (7) is standard Tikhonov regularization applied to the symmetry-reduced force model; the constrained construction is detailed in Appendix D. The acoustic sum rule is imposed as a hard linear constraint, and an optional central-force projection further reduces the parameter space. These constraints, the finite cutoff, and ridge regularization are physical and statistical priors, and their influence must be distinguished from information provided by the DFT forces.

For one n-atom configuration, $\mathbf{G}$ has $3n$ rows and hence algebraic rank no larger than $3n$. Full identification requires the rank to be compared with the number of independent IFC parameters after all symmetry and sum-rule reductions. Ridge regularization stabilizes a rank-deficient fit but does not create missing information. The validation therefore reports parameter counts, singular-value tolerance, rank, and condition number for each actual matrix rather than asserting coverage from the iteration count alone.

D. Phonon Reconstruction

From the extracted IFCs $\hat{\Phi}_{ij}^{\alpha\beta}(\mathbf{R})$, the dynamical matrix is constructed as:

$$D_{ij}^{\alpha\beta}(\mathbf{q}) = \frac{1}{\sqrt{M_i M_j}} \sum_{\mathbf{R}} \hat{\Phi}_{ij}^{\alpha\beta}(\mathbf{R})\, e^{i\mathbf{q}\cdot\mathbf{R}} \tag{8}$$

Equation (8) is the standard Fourier construction of the harmonic dynamical matrix from real-space IFCs [15]; the retained real-space cutoff and non-analytic polar correction determine its domain of validity.

Phonon dispersion: the $3n_b$ branches $\omega_{\mathbf{q}s}$ follow from eigenvalues of $D(\mathbf{q})$, evaluated on a dense Monkhorst–Pack grid. The phonon density of states (DOS) is obtained by linear tetrahedron integration with Gaussian broadening $\sigma = 0.1$ THz.

Thermodynamic properties follow directly from the phonon DOS $g(\omega)$:

$$C_V(T) = k_B \int_0^{\omega_{\max}} g(\omega) \left(\frac{\hbar\omega}{2k_BT}\right)^2 \sinh^{-2}\!\left(\frac{\hbar\omega}{2k_BT}\right) d\omega \tag{9}$$

$$F_{\mathrm{vib}}(T) = k_BT \int_0^{\omega_{\max}} g(\omega) \ln\!\left[2\sinh\!\left(\frac{\hbar\omega}{2k_BT}\right)\right] d\omega \tag{10}$$

$$\mathrm{ZPE} = \frac{\hbar}{2}\int_0^{\omega_{\max}} g(\omega)\,\omega\, d\omega \tag{11}$$

$$U_{\mathrm{iso}}(T) = \frac{\hbar}{6M}\int_0^{\omega_{\max}} \frac{g(\omega)}{\omega} \coth\!\left(\frac{\hbar\omega}{2k_BT}\right) d\omega, \qquad \int_0^{\omega_{\max}} g(\omega)d\omega = 3. \tag{12}$$

Equation (12) is the monatomic one-component displacement obtained from Eq. (3) when the total DOS is normalized to three modes per atom. For a compound, species-resolved eigenvector weights and masses are required; a species-averaged mass multiplying the total DOS is not used. The harmonic thermodynamic identities in Eqs. (9)–(12) are standard consequences of the independent-phonon partition function [15]. The spectral Debye temperature is the second-moment Debye temperature $T_D^{\mathrm{spec}} = (\hbar/k_B)\sqrt{5\langle\omega^2\rangle_g/3}$, where $\langle\omega^2\rangle_g = \int g(\omega)\,\omega^2\,d\omega / \int g(\omega)\,d\omega$ is the mean-square phonon frequency over the DOS; this uses the Debye relation $\langle\omega^2\rangle = 3\omega_D^2/5$. The caloric Debye temperature $T_D^{\mathrm{cal}}(T)$ is defined by inversion of the Debye model for $C_V(T)$.

E. Self-Consistent Loop

The SOTC method is closed into a self-consistency cycle:

1. **Initialize**: start from a Debye or Einstein trial phonon model $\omega^{(0)}(\mathbf{q})$.
2. **Compute target correlators**: $\bar{C}_2(R,T)$ from Eqs. (1)–(2) using current phonon model.
3. **Design configurations**: with the supercell fixed, minimize $\mathcal{Q}(\mathbf{u})$ [Eq. (4)] to generate N_{ens} displacement configurations.
4. **DFT forces**: evaluate the undisplaced reference and all N_{ens} displaced configurations.

5. **Extract IFCs**: ridge regression from Eq. (6).
6. **Update phonon model**: diagonalize $D(\mathbf{q})$ from new IFCs.
7. **Check convergence**: the archived benchmarks use $\|\Delta\bar{C}_2\|_2 < 3 \times 10^{-3}$ Å^2, together with the material-specific minimum iteration counts reported in the calculation inputs. If not converged, return to step 2.

The Debye or Einstein model supplies only the first target in the intended self-consistent workflow; subsequent targets depend on the fitted material-specific IFCs. Nevertheless, the finite-data priors and numerical safeguards can retain initialization dependence. Validation requires comparison of Debye, Einstein, perturbed, and deliberately poor initial models rather than assuming a unique fixed point.

Each iteration evaluates N_{ens} displaced configurations and an undisplaced reference configuration. In the current implementation the reference is recomputed once per iteration; hence the force-call count is $N_{\mathrm{iter}}(N_{\mathrm{ens}}+1)$, not N_{iter}. The archived benchmarks reach their stopping criterion in 2–5 iterations with $N_{\mathrm{ens}} = 1$–8, as detailed in Table I. Iteration count is therefore not used as a proxy for DFT cost.

F. Quasi-Harmonic Approximation and Continuous High-Temperature Properties

The self-consistent SOTC loop delivers converged IFCs $\Phi_{ij}^{*\alpha\beta}(T)$ at a single design temperature T. To obtain continuous thermodynamic properties, specifically the thermal expansion coefficient $\alpha(T)$, isobaric heat capacity $C_P(T)$, and bulk modulus $B(T)$, across a temperature range without repeating the SOTC loop at each T, we couple SOTC phonons to the QHA.

SOTC-QHA setup. A converged QHA calculation requires a set of N_V volumes spanning the thermally sampled range. Static reference-lattice energies $E_0(V)$ and phonons are computed directly at every volume; energies of thermally displaced configurations are not substituted for E_0, and IFCs at an uncalculated volume are not inferred by linear extrapolation. At least five volumes are used for a third-order Birch–Murnaghan fit, with convergence checked against the number and range of volumes. The zero-pressure Helmholtz free energy is

$$F(T,V) = E_0(V) + F_{\mathrm{vib}}(T,V), \tag{13}$$

where the harmonic vibrational contribution is

$$F_{\mathrm{vib}}(T,V) = k_B T \sum_{\mathbf{q},s} \ln\left[2\sinh\left(\frac{\hbar\omega_{\mathbf{q}s}(V)}{2k_BT}\right)\right]. \tag{14}$$

At external pressure p, the equilibrium volume minimizes $F(T,V)+pV$:

$$\left.\frac{\partial[F(T,V)+pV]}{\partial V}\right|_{V^*(T,p)} = 0. \tag{15}$$

Thus F, rather than G, is minimized at $p = 0$. Equations (13)–(15) are the standard quasi-harmonic construction [16, 17]; they do not imply an anharmonic variational upper bound.

For a cubic crystal, define the volumetric and linear expansion coefficients separately. The thermodynamic identities are

$$\alpha_V = \frac{1}{V^*}\frac{dV^*}{dT}, \qquad \alpha_L = \frac{\alpha_V}{3}, \qquad B_T = V^*\left.\frac{\partial^2 F}{\partial V^2}\right|_{V^*}. \tag{16}$$

The corresponding heat-capacity relation is

$$C_P - C_V = TV^*B_T\alpha_V^2 = 9TV^*B_T\alpha_L^2. \tag{17}$$

The positive curvature sign, the distinction between α_V and α_L, and the factor of nine in the linear-coefficient form are essential. All numerical QHA results must be regenerated with these conventions and with a converged volume grid before an accuracy or cost comparison is made.

Grüneisen shortcut. When volume-changing DFT runs are not desired, a faster estimate uses the mode Grüneisen parameters:

$$\gamma_{\mathbf{q}s} = -\frac{V}{\omega_{\mathbf{q}s}}\left.\frac{\partial\omega_{\mathbf{q}s}}{\partial V}\right|_{V_0} \approx -\frac{V_0}{\omega_{\mathbf{q}s}}\frac{\omega_{\mathbf{q}s}(V_+)-\omega_{\mathbf{q}s}(V_-)}{V_+-V_-} \tag{18}$$

The heat-capacity-weighted average $\gamma_{\mathrm{eff}}(T) = \sum_{\mathbf{q}s} c_{\mathbf{q}s}(T)\gamma_{\mathbf{q}s} / \sum_{\mathbf{q}s} c_{\mathbf{q}s}(T)$ gives:

$$\alpha_L(T) \approx \frac{\gamma_{\mathrm{eff}}(T)\, C_V(T)}{3B_0V_0}, \qquad \alpha_V(T) = 3\alpha_L(T). \tag{19}$$

Here $B_0 = V_0(\partial^2E_0/\partial V^2)_{V_0}$ is the positive static bulk modulus. Equation (19) is the standard cubic Grüneisen relation [16]; it requires volume derivatives of the phonons and is a diagnostic approximation, not an independent validation of QHA. Software-specific class names and keyword choices are moved to the reproducibility material.

III. COMPUTATIONAL DETAILS

A. DFT Calculations

All force calculations used Quantum ESPRESSO 7.5 [18], PBE [19], and pseudopotentials from the SSSP efficiency library [20]. The selected potentials were ONCV for Ag, Au, and Mo; PAW for Al, Cu, Mg, O, and Pb; and ultrasoft for Na, Cl, Te, and W. Wave-function cutoffs (Ry) were Al 40, Ag 50, Au 45, Cu 45, Mo 40, W 30, NaCl 60, MgO 80, and PbTe 44. Charge-density cutoffs were four times the wave-function cutoff for ONCV and eight times it for PAW/ultrasoft potentials. FCC and BCC supercells used a Gamma-centered $2 \times 2 \times 2$ k

mesh; rocksalt supercells used Gamma only. Metals used Marzari–Vanderbilt cold smearing [21] of 0.02 Ry, while the insulators used fixed occupations. Literal pseudopotential filenames, QE keywords, and software class/flag names are collected in Appendix K rather than the theory section.

Experimental lattice constants and ideal high-symmetry fractional coordinates were used for the archived benchmarks. An undisplaced single-point calculation at the same cell defines the reference force and stress. IFC fits use $\Delta\mathbf{F} = \mathbf{F}(\mathbf{u}) - \mathbf{F}(\mathbf{0})$, as in Eq. (6), rather than assuming $\mathbf{F}(\mathbf{0}) = 0$. The calculation archive should report maximum/RMS reference forces, induced-force scales, residual stresses, and comparisons with representative PBE-relaxed cells.

B. SOTC Supercell Setup

The supercell geometry (Hermite Normal Form matrix $\mathbf{H}$) is selected once before the self-consistency loop by enumerating all HNF candidates whose number of primitive cells n satisfies $\max(2,\, n_{\mathrm{sc}} - 2) \le n \le n_{\mathrm{sc}}$, where n_{sc} is the user-specified target. Candidates are ranked by a two-criterion score: nearest-neighbor shell completeness first, then supercell lattice condition number (a measure of cell cubicity); the best candidate at or below the target size is adopted. This upper bound ensures the supercell never exceeds the intended size.

FCC metals (Al, Ag, Au, Cu) used a $2 \times 2 \times 2$ conventional-cell supercell (4 atoms/cell $\times$ 8 = 32 atoms); BCC metals (Mo, W) used a 64-atom supercell (2 atoms/cell $\times$ 32 conventional cells) and rocksalt compounds (NaCl, MgO, PbTe) used a $2\times2\times2$ conventional-cell supercell (8 atoms/cell $\times$ 8 = 64 atoms).

The experimental lattice parameters are provided in the archived inputs; Table I now reports force-call accounting. The IFC cutoffs are listed in Table I, and the ridge parameter was 10^{-3} in the code's reported units. A central-force bond projection was enabled for all archived benchmarks. Because this is a strong physical prior rather than a software detail, sensitivity tests should compare it with the less constrained tensor fit.

The archived benchmark workflow used one SOTC displacement configuration per self-consistency iteration ($N_{\mathrm{ens}} = 1$). This fact does not guarantee full IFC coverage. For every benchmark, the data release should report the dimensions and numerical rank of the symmetry-reduced design matrix, its singular-value threshold and condition number, and the independent parameter count. The ensemble mode ($N_{\mathrm{ens}} > 1$) provides additional force rows when an overdetermined fit is required.

C. Phonon Post-Processing

Phonon DOS calculations used a $30 \times 30 \times 30$ Monkhorst–Pack mesh (27,000 q points), while band dispersions were interpolated from the real-space IFCs along the paths stored with the released band data: Γ-X-W-K-Γ-L for FCC and rocksalt and Γ-H-N-Γ-P-H for BCC. Thermal properties were computed on an $8 \times 8 \times 8$ grid (512 q-points) for $C_V(T)$, $F_{\mathrm{vib}}(T)$, and $S_{\mathrm{vib}}(T)$. Gaussian broadening $\sigma = 0.1$ THz was applied to the phonon DOS. Convergence of C_V with q-mesh was verified by comparing $8 \times 8 \times 8$ and $16 \times 16 \times 16$ grids for Al and MgO; the difference is < 0.05 J/(mol·K) in both cases, confirming that the $8\times8\times8$ grid is fully converged.

D. Experimental Reference Data

Experimental Debye temperatures and heat capacities were taken from the NIST-JANAF tables [22] and Kittel [23]. References for individual systems: Al [24, 25], Ag [25], Au [26], Cu [27], Mo [28, 29], W [29], NaCl [30], MgO [31, 32], PbTe [33, 34]. For comparison of phonon dispersions, we use neutron scattering data from: Cu (Ref. [35]), Al (Ref. [36]), NaCl (Ref. [37]).

E. SOTC-QHA Setup for Aluminum

The available Al dataset contains only V_0 and V_-, an extrapolated expanded-volume IFC matrix, and energies from displaced cells. It is insufficient for a converged equation of state. A quantitative QHA study requires direct static reference-lattice energies and direct FC2 phonon calculations at at least five volumes, followed by volume-grid, EOS, and temperature-derivative convergence tests as specified in Sec. II.F. No numerical Al expansion coefficient or QHA cost/accuracy comparison is therefore reported.

IV. RESULTS

A. Self-Consistency Convergence

A multi-method Debye-temperature comparison is not shown because the available external-method series lack calculation-level provenance and matched settings. The SOTC FC2 and experimental values remain tabulated, but no external-method accuracy claim is based on those unmatched data.

The archived runs stop after 2–5 self-consistency iterations under the material-specific minimum-iteration rule and the correlator threshold stated in Sec. II.E. This is convergence of the chosen scalar stopping metric, not a demonstration that the full phonon dispersion is converged. Direct configuration-count and dispersion-error convergence against a matched reference is required for that stronger claim.

Table I shows that the present runs themselves use ensembles and require 2–32 displaced calculations, or 4–36 force calculations when the repeated reference evalu-

TABLE I. Archived SOTC iteration and force-call accounting. N_{iter} is the number of self-consistency iterations, N_{ens} the displaced configurations evaluated per iteration, $N_{\mathrm{disp}} = N_{\mathrm{iter}} N_{\mathrm{ens}}$, and $N_{\mathrm{force}} = N_{\mathrm{iter}}(N_{\mathrm{ens}} + 1)$ includes the undisplaced reference recomputed by the current runner. These counts do not establish IFC or dispersion convergence.

Material	T_{des} (K)	N_{atoms}	r_{cut} (Å)	N_{iter}	N_{ens}	N_{disp}	N_{ref}	N_{force}
Al	300	32	4.5	2	5	10	2	12
Ag	1000	32	4.5	4	5	20	4	24
Au	1200	32	4.5	2	1	2	2	4
Cu	1200	32	4.5	4	8	32	4	36
Mo	1500	64	5.0	2	1	2	2	4
W	1500	64	5.0	4	3	12	4	16
NaCl	700	64	4.3	5	3	15	5	20
MgO	1500	64	3.2	4	3	12	4	16
PbTe	700	64	5.0	4	5	20	4	24

ations are counted. No external-method cost-reduction factor is inferred from these numbers because no comparison dataset was generated with a matched cell, electronic-structure setup, sampling protocol, IFC rank, and dispersion accuracy target.

B. Phonon Dispersions and DOS

Figure 2 displays the SOTC effective-FC2 dispersions and DOS. The curves expose branch topology and instabilities but do not include explicit FC3 or FC4 self-energies. They are presented descriptively because the cited neutron data and a matched converged DFT reference are not overlaid; no branch-resolved accuracy or superiority claim is inferred.

The computed branch counts, acoustic/optic separation, and overall frequency ranges are qualitatively consistent with neutron-scattering dispersions for Al, Cu, and NaCl [35–37]. The Al calculation and neutron data are both available at 300 K, whereas the Cu and NaCl comparisons involve unmatched temperatures. A quantitative comparison requires digitized points on identical paths and temperatures, followed by frequency MAE/RMSE, maximum branch error, sound-velocity error, and instability fraction versus configuration count. The NaCl zone-center optic comparison is quantified separately below. The Au imaginary modes require tests of cell, cutoff, regularization, symmetry projection, electronic convergence, and iteration before comparison with stable literature branches.

For NaCl, the non-analytic polar correction was evaluated explicitly. Primitive-cell QE 7.5 DFPT calculations used PBE, the same Na and Cl pseudopotentials as the force calculations, 60/480 Ry wave-function/charge cutoffs, an $8 \times 8 \times 8$ electronic mesh, 10^{-10} Ry SCF convergence, and 10^{-14} DFPT convergence. The charge-neutrality-projected tensors are $Z^*_{\mathrm{Na}} = +1.110915\,\mathbf{I}$, $Z^*_{\mathrm{Cl}} = -1.110915\,\mathbf{I}$, and $\varepsilon_\infty = 2.470969\,\mathbf{I}$. Applying these tensors through the phonopy non-analytic correction to the matched central finite-displacement QE FC2 force constants changes the directional Γ-point optic limit from 4.369 (TO) to 7.264 THz (LO), a 2.895 THz splitting (Fig. 2).

Archival infrared dielectric data in NBS Special Publication 301 report zone-center TO and LO limits of 164 and 268 cm^{-1}, respectively [38]. These correspond to 4.917 and 8.034 THz and are overlaid in Fig. 2. Thus the present QE harmonic calculation is lower by 0.547 THz (11.1%) for TO and 0.770 THz (9.6%) for LO; its splitting is 0.223 THz (7.2%) below the 3.118 THz optical value. In contrast, the calculated LO/TO ratio, 1.663, is only 1.7% above the optical ratio of 1.634. The similar absolute softening of both modes, together with the much smaller ratio error, suggests a common frequency-scale contribution from the chosen functional, lattice parameter, and unmatched temperature, but does not by itself identify which contribution dominates. Room-temperature IXS and SSCHA calculations further show that the NaCl LO spectral function is strongly anharmonic and can contain multiple peaks along Γ–L [39]. Consequently, the present fixed-cell harmonic-NAC curve is compared with literature only at the optical Γ limits; it is not assigned a finite-temperature dispersion RMSE.

The primary nine-panel SOTC figure remains an archived no-NAC result and is labeled accordingly. Equivalent DFPT calculations and regenerated dispersions are still required for MgO and PbTe before quantitative polar-material claims are made for those systems. The NaCl plot and machine-readable arrays are generated by `codes/validation/plot_nacl_loto.py`.

C. Heat Capacities

Table II reports the effective-harmonic C_V values obtained by integrating the SOTC DOS through Eq. (9), together with the cited calorimetric values. These fixed-volume results use one effective-FC2 model selected at T_{des}; they do not include thermal expansion, electronic heat capacity, or explicit higher-order anharmonic contributions. Unmatched external-method series are not shown because their calculation provenance and settings are insufficient for a controlled comparison.

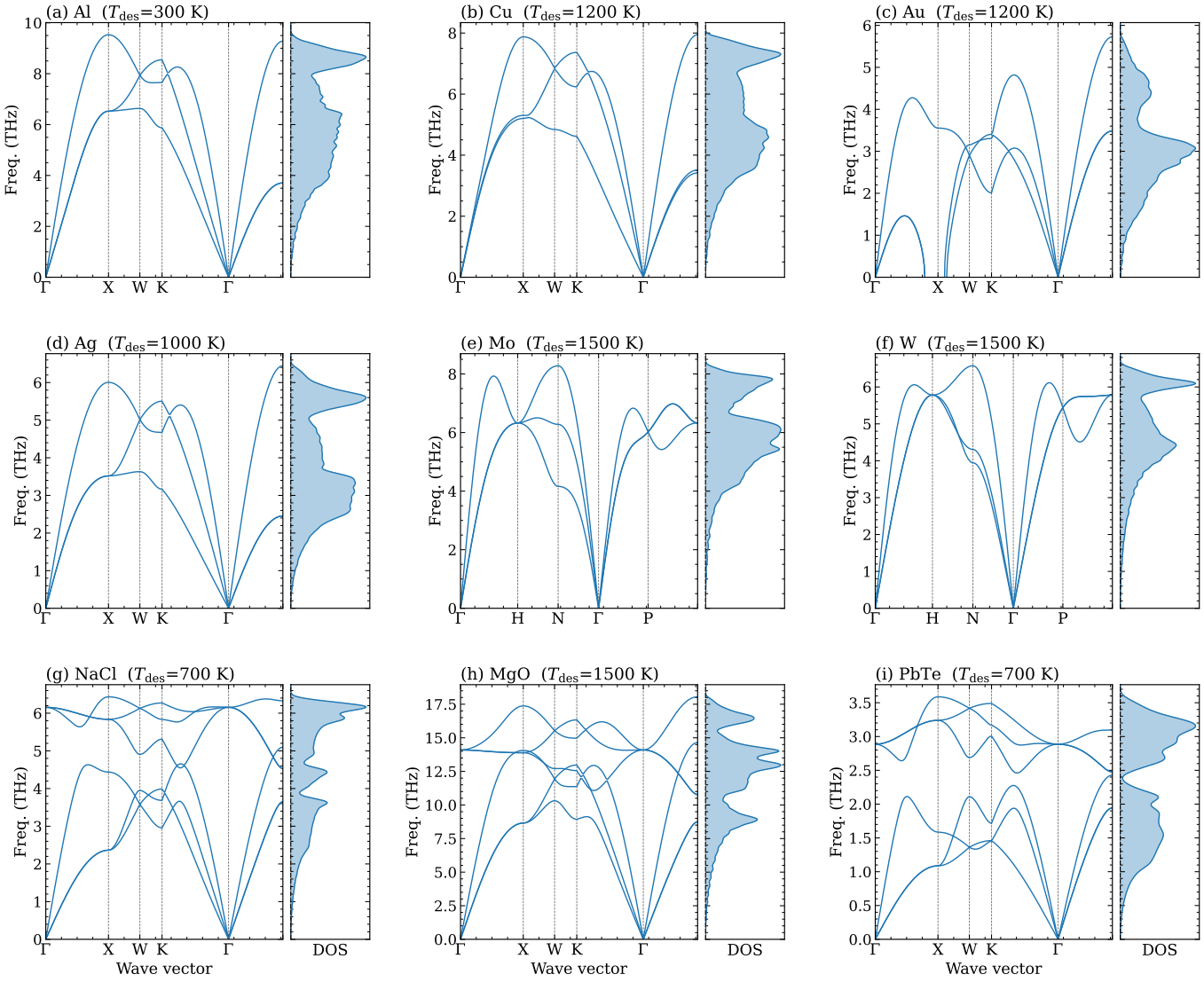


FIG. 1. SOTC effective-FC2 phonon dispersions and DOS for (a) Al (300 K), (b) Cu (1200 K), (c) Au (1200 K), (d) Ag (1000 K), (e) Mo (1500 K), (f) W (1500 K), (g) NaCl (700 K), (h) MgO (1500 K), and (i) PbTe (700 K). The stored band paths are Γ-X-W-K-Γ-L for FCC and rocksalt and Γ-H-N-Γ-P-H for BCC. DOS data use a $30 \times 30 \times 30$ q mesh and 0.1 THz broadening. Experimental neutron points are not overlaid in this archived figure. The three polar materials omit the non-analytic LO–TO correction, and the Au panel contains an instability; these omissions preclude a quantitative accuracy claim.

T_D^{SOTC} is the second-moment spectral value from the postprocessed IFC dataset, C_V^{SOTC} is evaluated at T_{des}, and ZPE is reported per formula-unit atom. Spectral, caloric, and ultrasonic Debye temperatures are not interchangeable; their differing definitions and measurement temperatures limit a point-by-point comparison.

The archived runner summaries and the postprocessed thermal summaries do not contain identical T_D values for every material. Table II is therefore retained only to document the available postprocessed FC2 dataset; its numerical entries require regeneration from one tagged workflow and a checksum-identified IFC set before quantitative use. No accuracy statistic is assigned to the table in its present form.

At high temperature the harmonic phonon expression approaches $3R$, whereas experimental heat capacities can include electronic and explicit anharmonic terms. Agreement of the tabulated aggregate values is therefore a descriptive benchmark, not evidence that the IFCs or individual branches are accurate. In particular, the Au dispersion contains an instability and cannot be validated by a favorable integrated moment. Direct branch-resolved reference calculations are required before the material-by-material accuracy statements are made.

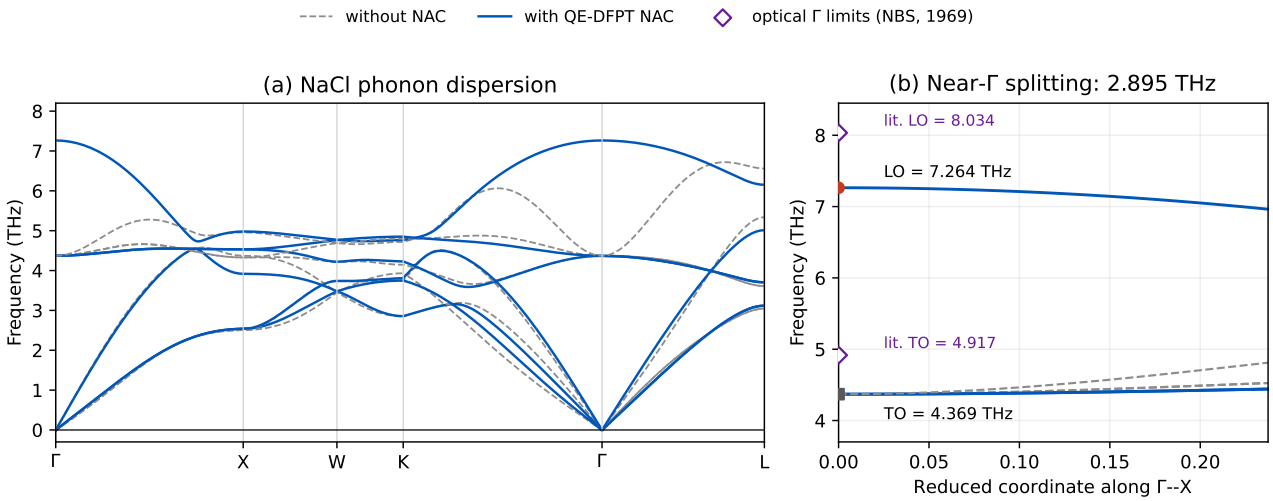


FIG. 2. NaCl LO–TO validation using the completed QE DFPT Born effective charges and electronic dielectric tensor. (a) Matched central-finite-displacement phonopy FC2 dispersion without the non-analytic correction (gray dashed) and with the QE-DFPT correction (blue) along Γ-X-W-K-Γ-L. (b) Near-Γ optic branches along Γ-X. The transverse limit is 4.369 THz and the corrected longitudinal limit is 7.264 THz, giving a 2.895 THz splitting. Purple diamonds show the archival optical Γ limits of 4.917 and 8.034 THz [38]; the calculated TO, LO, and splitting are lower by 11.1%, 9.6%, and 7.2%, respectively. The figure uses the conventional finite-displacement reference so that the polar correction is isolated from the SOTC sampling question.

TABLE II. Benchmark summary: computed vs. experimental Debye temperatures and heat capacities.

Material	Struct.	T_{des}	$\omega_{\mathrm{max}}^{\mathrm{SOTC}}$ (THz)	T_D^{SOTC} (K)	T_D^{expt} (K)	C_V^{SOTC} (J/mol·K)	C_V^{expt} (J/mol·K)	ΔC_V (%)	ZPE (eV/atom)
Al	FCC	300 K	10.3	450	390–428 [24]	22.9	24.4	−6.1	0.038
Ag	FCC	1000 K	6.8	297	∼225 [25]	24.9	27.1	−8.3	0.024
Au	FCC	1200 K	6.5	282	165† [26]	24.9	27.1	−8.1	0.020
Cu	FCC	1200 K	9.2	401	315–343 [27]	24.8	26.4	−5.9	0.034
Mo	BCC	1500 K	8.7	380	380–470 [28, 29]	24.9	28.7	−13.3	0.035
W	BCC	1500 K	7.3	318	310–400 [29]	24.9	26.9	−7.5	0.029
NaCl	Rocksalt	700 K	7.2	300	321 [30]	49.5	51.7	−4.4	0.056
MgO	Rocksalt	1500 K	20.3	726	743–800 [31, 32]	49.2	49.2	0.0	0.153
PbTe	Rocksalt	700 K	4.2	128	130–160 [33, 34]	47.3	48.3	−2.2	0.025

D. Zero-Point Energies and Vibrational Free Energies

The zero-point energies (ZPE) and vibrational free energies in Table II are computed from the converged phonon DOS via Eqs. (10)–(11) (harmonic phonon formulas at the equilibrium volume V_0).

The zero-point energies (Table II) range from 0.020 eV/atom (Au, heavy mass, soft phonons) to 0.153 eV/atom (MgO, light mass, hard phonons), fully consistent with the ZPE $\propto M^{-1/2}\omega$ scaling. The ZPE of Al (0.038 eV/atom) is in agreement with the ZPVE-corrected DFT value of 0.038–0.041 eV/atom from Refs. [24, 25].

The vibrational entropy $S_{\mathrm{vib}}(T)$ at 1000 K spans 57–122 J/mol·K across the benchmark set, in the expected order (soft/heavy PbTe, Au > NaCl > Cu > Mo > Al > hard MgO). For aluminum, $S_{\mathrm{vib}}^{\mathrm{SOTC}}(1000\,\mathrm{K}) = 57$ J/(mol·K) agrees to within 1% with the value obtained by integrating the NIST-JANAF heat capacity $\int_0^{1000}(C_V/T')\,dT'$ [22], confirming that the SOTC phonon spectrum accurately captures the full spectral weight across the Brillouin zone.

E. Mean-Square Displacements

Equations (3) and (12) define the same one-component isotropic displacement parameter U_{iso}, with $B = 8\pi^2 U_{\mathrm{iso}}$. The archived Al, Au, and Cu values have an ambiguous total-DOS normalization and are not compared with experiment. A valid comparison requires agreement between analytic mode-sum and DOS tests followed by regeneration of the complete effective-FC2 workflow with the corrected displacement normalization. Consequently, the present benchmarks do not validate Debye–Waller factors.

F. Thermal Expansion and Isobaric Heat Capacity (QHA)

No Al QHA curve or table is reported because the available two-volume FC2 dataset uses an extrapolated expanded-volume force-constant matrix and produces a nonmonotonic 300 K linear expansion coefficient. It also mixes linear and volumetric expansion conventions and an inconsistent curvature sign, so it cannot support an accuracy or DFT-cost comparison.

A quantitative QHA benchmark requires direct FC2 phonon calculations on a converged volume grid, static reference-lattice energies, the identities in Eqs. (13)–(17), and convergence with respect to EOS choice, volume range, temperature grid, and numerical differentiation. Full-QHA and Grüneisen results should then be tabulated from the same regenerated dataset.

G. Independent Higher-Order IFC Validation

We additionally tested whether the software can recover explicit higher-order force content in a workflow independent of the few-configuration SOTC benchmarks. Matched calculations were performed for FCC Al at 300 K, BCC W at 1500 K, and rocksalt NaCl at 700 K in $3 \times 3 \times 3$ primitive-cell supercells containing 27, 27, and 54 atoms, respectively. For each material, 50 configurations drawn from the quantum harmonic canonical distribution were paired with their exact inversions, giving 100 snapshots. Complete $+\mathbf{u}/-\mathbf{u}$ pairs were kept wholly within an 80-snapshot training set or a strictly held-out 20-snapshot test set.

As a conventional harmonic baseline, central finite-displacement FC2 tensors were generated with phonopy from matched QE calculations in the same supercells, using +0.01 and –0.01 Å displacements and explicit subtraction of the undisplaced reference force. The thermal snapshots were then fit with the same symmetry-reduced `symfc` design, successively retaining FC2, FC2+FC3, and FC2+FC3+FC4 terms. Space-group, index-permutation, translation, and acoustic-sum-rule constraints were imposed without ridge regularization. Table III evaluates every model on the unchanged test set. These are predictive force tests rather than comparisons of training residuals.

Temperature-fitting FC2 alone gives only modest improvements over the finite-displacement harmonic baseline: 3.3% for Al, 6.4% for NaCl, and 7.5% for W. Adding FC3 lowers the T-fit FC2 error by 72.7%, 43.0%, and 68.7%, respectively; adding FC4 then lowers the remaining error by 76.1%, 48.1%, and 56.8%. Relative to the conventional FD FC2 baseline, the final reductions are 93.7%, 72.3%, and 87.5%. The final held-out R^2 values are 0.99974, 0.96907, and 0.99943. All normalized designs have full column rank (condition numbers 312.7, 36.9, and 269.4), and the largest compact acoustic-sum-rule residual is 1.2×10^{-12} in the tensor's native units.

Snapshot-count fits using 20, 40, 60, and 80 training configurations, cutoff brackets of ± 0.5 Å, and selected plane-wave-cutoff and k-mesh calculations were archived. The Al cutoff test selected 5.0 Å for FC2/FC3: relative to 4.5 Å, its held-out RMSE falls from 0.03103 to 0.00651 eV Å^{-1}. The denser-k-mesh force shift for Al remains larger than this final fit error, so the Al tensors are not presented as electronically converged transport coefficients. NaCl has the largest sampled displacement amplitudes and the weakest held-out score; its result establishes substantial higher-order force content but not converged transport properties.

The production archive contains compact FC2/FC3/FC4 HDF5/NumPy tensors, phono3py `fc3.hdf5`, ShengBTE `FORCE_CONSTANTS_3RD`, and FourPhonon `FORCE_CONSTANTS_4TH`. Tensor conventions are covered by synthetic cross-format tests. Table III is regenerated from the archived JSON, NPZ, and HDF5 files by `codes/validation/summarize_real_anharmonic_ifcs.py`; no table entry is transcribed from terminal output. No linewidth, lifetime, lattice thermal conductivity, or higher-order free-energy result is inferred here. Most importantly, this 100-snapshot canonical campaign validates the higher-order fitting and export path; it does not establish a sampling advantage for the few-configuration SOTC design.

V. DISCUSSION

A. Physical Validity of the Effective Harmonic Approximation

The primary SOTC self-consistency workflow fits a second-order effective-harmonic model. It can represent temperature-dependent curvature only insofar as the DFT force data and self-consistency procedure encode it in the fitted IFCs. The independent workflow in Sec. IV G fits and exports explicit FC3 and FC4 tensors from a much larger canonical ensemble, but no linewidth, lifetime, transport coefficient, non-Gaussian nuclear distribution, or explicitly anharmonic free energy is computed here. Consequently, agreement of C_V, ZPE, or a Debye moment with experiment cannot by itself establish an accurate branch-resolved dispersion or an anharmonic variational solution: these are DOS-integrated observables that can hide compensating frequency errors.

The archived benchmarks show that the scalar correlator stopping metric becomes smaller than the chosen threshold in 2–5 iterations. That observation is limited to empirical convergence of the selected scalar metric. The finite fit also relies on the IFC cutoff, space-group and central-force constraints, the acoustic sum rule, ridge regularization, the initialization, and 1–8 configurations per iteration. Their separate contributions have not yet been isolated. Matched random/Gaussian controls, initialization ablations, SVD diagnostics, and direct dispersion

TABLE III. Matched model comparison for real-material higher-order IFC validation. RMSE is the Cartesian force-component error in eV Å^{-1} on 20 strictly held-out canonical snapshots. *FD FC2* is the conventional central finite-displacement phonopy model; *T-fit* models are fit to the 80 thermal training snapshots. p/r is the number of symmetry-reduced parameters and normalized design rank for the final FC2+FC3+FC4 fit. The final column is the RMSE reduction relative to FD FC2.

Material	T (K)	atoms	p/r	FD FC2	T-fit FC2	T-fit FC2+FC3	T-fit FC2+FC3+FC4	reduction (%)
Al	300	27	106/106	0.10294	0.09955	0.02721	0.00651	93.7
NaCl	700	54	48/48	0.35432	0.33152	0.18906	0.09805	72.3
W	1500	27	126/126	0.29816	0.27566	0.08634	0.03730	87.5

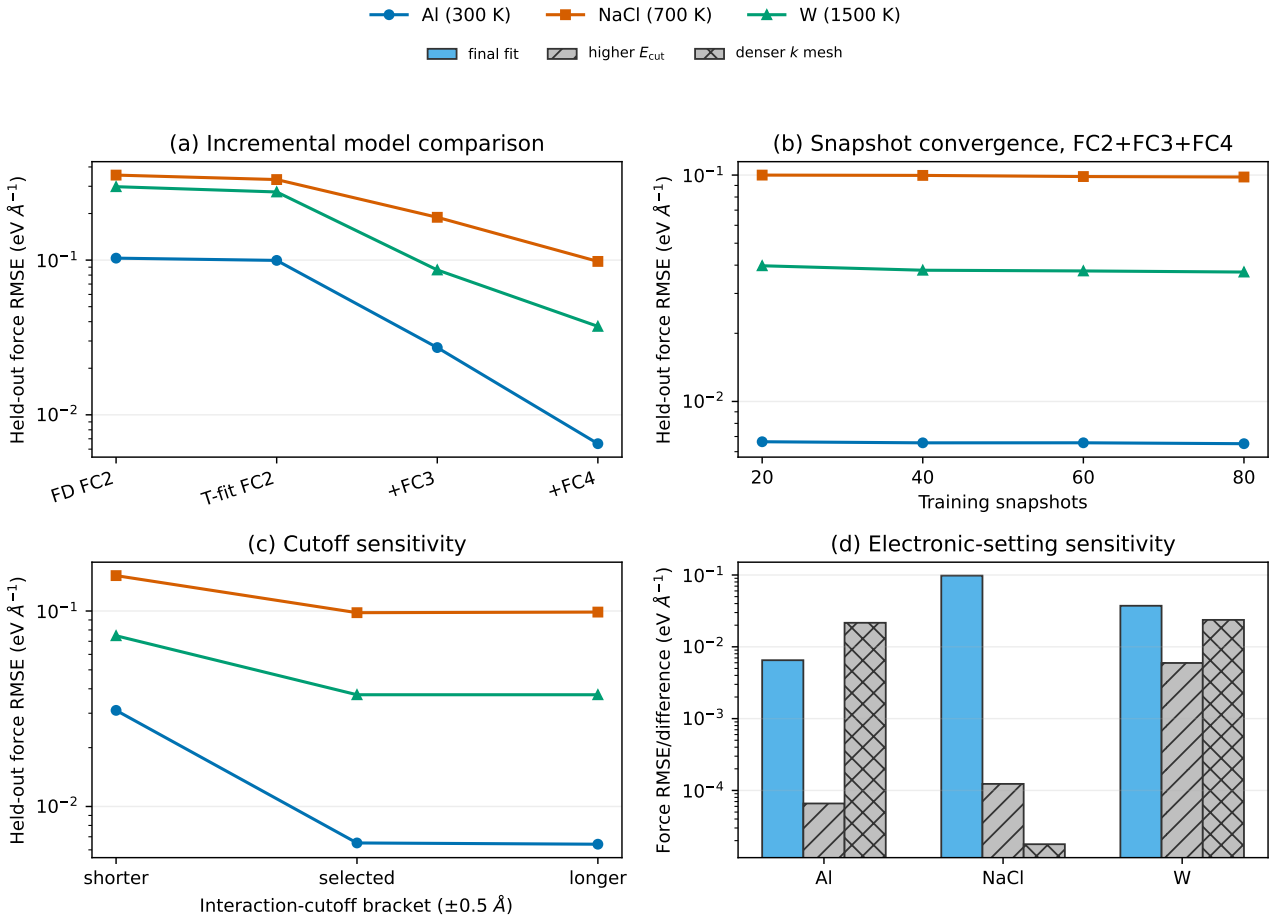


FIG. 3. Post-IFC validation for the completed QE force campaign. (a) Held-out force-component RMSE for conventional finite-displacement FC2, temperature-fitted FC2, FC2+FC3, and FC2+FC3+FC4 models. (b) Final-model convergence with the number of thermal training snapshots. (c) Sensitivity to interaction cutoffs bracketed by ±0.5 Å around the selected values. (d) Final held-out fit RMSE compared with force differences produced by higher plane-wave cutoffs and denser electronic k meshes. Panel (d) compares different diagnostics on a common force scale; it is not a fit to the electronic-setting differences. All predictive errors use the unchanged 20-snapshot test set. Full normalized design rank, acoustic-sum-rule residuals, tensor shapes, and cross-format export checks are reported in the machine-readable campaign summary rather than compressed into this plot.

convergence against fully converged DFT are therefore required before attributing any improvement specifically to correlator optimization.

The Au data require particular caution. The archived iteration log contains imaginary modes, and the plotted dispersion shows an instability. Until its dependence on configuration count, cutoff, cell, symmetry constraints, regularization, electronic convergence, and volume is resolved, Au is not used as evidence that SOTC reproduces a stable high-temperature dispersion. Similar caution applies to the polar compounds because the current dynamical matrices omit the non-analytic LO–TO correction.

B. Computational Efficiency

The relevant cost is the number and size of electronic-structure force calculations, not the number of outer self-consistency iterations. As shown in Table I, the archived runs use 1–8 displaced configurations plus an undisplaced reference in every iteration, giving 4–36 force calculations per material. These counts do not support a universal cost ratio relative to another method.

A defensible efficiency comparison must use the same material, cell and k-point convergence, exchange-correlation approximation, physical observable, and quantitative accuracy threshold. It must also count reference forces, discarded/restarted calculations, QHA volumes, and DFPT non-analytic corrections. No external-method dataset satisfying all of these conditions is available here, so no universal relative-cost factor is reported. Wall-time and core-hour comparisons require matched, reproducible calculations.

C. Comparison with Related Methods

The temperature-dependent effective potential (TDEP) framework fits effective second- and higher-order IFCs to finite-temperature force data; molecular dynamics, Gaussian sampling, and other ensembles can supply those data [5–8]. The stochastic self-consistent harmonic approximation (SSCHA) instead optimizes a variational harmonic trial Hamiltonian and can incorporate strong quantum and thermal anharmonic effects in the free energy and renormalized spectrum [9–11]. Mode-coupling approaches provide a distinct non-perturbative route to renormalized lattice dynamics [12], while AIMD generates classical nuclear trajectories on a first-principles potential-energy surface [13].

In IFC-rank terms, the primary SOTC workflow and every dispersion in Fig. 1 use an effective FC2 model; the conventional phonopy and LO–TO control in Fig. 2 also use FC2. The independent campaign in Fig. 3 explicitly compares FC2, FC2+FC3, and FC2+FC3+FC4 fits. The cited TDEP literature includes effective FC2 treatments [5, 6] and an explicit effective-FC3 formulation [8]. SSCHA's optimized trial Hamiltonian is harmonic (FC2), although its free-energy derivatives and spectra contain anharmonic effects; it should not be described as an explicit FC3/FC4 tensor fit. Mode-coupling theory is likewise not classified by a single fitted IFC rank.

SOTC has a narrower objective: it optimizes a finite set of spatial displacement moments before evaluating forces and then fits effective FC2. Its sampling advantage must be assessed with matched controls at the same physical level, cell, temperature, electronic settings, and target observable. The present benchmarks do not compare explicit anharmonic free energies or renormalized spectra with the established methods above. The special displacement method [40] is a particularly relevant deterministic comparison.

D. Limitations and Planned Extensions

LO-TO splitting. Polar insulators (NaCl, MgO, PbTe) require a non-analytic correction to the dynamical matrix [41]:

$$D_{ij}^{\alpha\beta,\mathrm{NAC}}(\hat{\mathbf{q}}) = \frac{4\pi e^2}{\Omega_0} \frac{(\sum_\gamma Z_i^{*\alpha\gamma}\hat{q}_\gamma)(\sum_\delta Z_j^{*\beta\delta}\hat{q}_\delta)}{\sum_{\gamma\delta}\hat{q}_\gamma\varepsilon_\infty^{\gamma\delta}\hat{q}_\delta}$$

Here Ω_0, Z^*, and ε_∞ are the primitive-cell volume, Born effective-charge tensor, and clamped-ion dielectric tensor, respectively. Figure 2 now applies real QE DFPT tensors to the matched NaCl finite-displacement phonons and resolves the missing longitudinal branch. The archived nine-panel SOTC figure itself is retained without NAC for provenance rather than silently replaced. MgO and PbTe still lack corresponding DFPT tensors and regenerated dispersions, so the quantitative polar-material accuracy claim remains restricted to the explicit NaCl LO–TO test.

Higher-order scope. The canonical-ensemble extension in Sec. IV G validates constrained FC3 and FC4 fitting and common transport-code file conventions. Phonon lifetimes and lattice thermal conductivity additionally require converged reciprocal- space integration, scattering phase space, isotope/boundary choices, and transport calculations [42]; none is reported here. No universal 5–10-configuration estimate is made: the demonstrated fits use 100 snapshots, and the required number depends on parameter count, symmetry, cutoff, matrix rank, force noise, and target accuracy.

Dynamically unstable systems. For systems near structural phase transitions ($BaTiO_3$, SnSe, BCC Ti below ∼900 K), the harmonic trial correlators in Eq. (1) cannot describe the anharmonic double-well landscape. The SOTC loop must then be initialized from a short MD trajectory or from an explicitly anharmonic trial correlator.

Quasi-harmonic and magnetic limitations. QHA neglects explicit temperature-dependent phonon–phonon terms at fixed volume and requires a converged volume dependence of both static energy and phonons. The available Al dataset does not demonstrate that these errors or the interpolation error are minor. Strongly anharmonic, dynamically unstable, or high-temperature systems require validation beyond QHA. Magnetic materials additionally require a treatment of spin–lattice coupling not included here [43].

E. SOTC-QHA vs. Phonopy-QHA

Phonopy is an open-source lattice-dynamics package that can organize finite-displacement FC2 phonon and QHA workflows [17]. A controlled comparison requires documented displacement settings and cannot combine an assumed displacement count with an unvalidated $N^{3/2}$

cost model or an underconverged SOTC-QHA dataset. Both workflows must report the code version, functional, pseudopotentials, cell, k/q meshes, displacement count, volume grid, convergence targets, wall time, and core-hours.

VI. CONCLUSIONS

SOTC is a deterministic design strategy for collective displacement configurations used in a symmetry-constrained, ridge-regularized fit of second-order IFCs. The archived nine-material calculations reach their chosen shell-correlator stopping criterion in 2–5 outer iterations, using 1–8 displaced configurations per iteration. Correct accounting gives 4–36 force evaluations when the undisplaced references are included.

A separate 100-snapshot canonical campaign establishes that the symmetry-constrained higher-order implementation produces full-rank FC2/FC3/FC4 fits for Al, NaCl, and W. Relative to matched central finite-displacement phonopy FC2 baselines, the final models lower held-out force RMSE by 72–94% and satisfy the acoustic sum rule to 1.2×10^{-12} or better. Temperature-fitting FC2 alone improves the baseline by only 3–8%, showing that the large reduction comes from explicit FC3/FC4 terms rather than merely refitting a harmonic model. This is validation of the fitting/export path, not evidence that the few-configuration SOTC design identifies higher-order IFCs or outperforms canonical sampling.

The primary SOTC evidence concerns effective-FC2 DOS-derived observables; it does not establish branch-resolved dispersion convergence, universal representability, or superiority over matched random/Gaussian configurations. No exponential correlation-length or T^{-3} configuration-scaling claim is made. Quantitative Al QHA and Debye–Waller results require corrected definitions and converged recalculation. Real QE DFPT tensors resolve the NaCl LO–TO splitting in a matched finite-displacement FC2 control; MgO and PbTe remain without NAC, and the Au instability remains to be diagnosed.

The next validation steps are therefore concrete: report the complete parameter/rank/SVD diagnostics, compare against matched random and converged finite-displacement controls, overlay neutron data, resolve the Au instability, complete DFPT/NAC dispersions for MgO and PbTe, rerun the MSD calculation with a single convention, and recompute QHA on a converged volume grid. Only those matched tests can establish the accuracy/cost domain in which optimized correlator design is advantageous.

ACKNOWLEDGMENTS

All calculations were performed on a local workstation server without external computational resources or funding support. The author declares no competing interests.

DATA AND SOFTWARE AVAILABILITY

The code, benchmark inputs, and archived outputs are available at `https://github.com/albert-hzbn/sotc`. The repository contains the Python implementation, Quantum ESPRESSO inputs, benchmark runners, fitted IFC and phonon data, and JSON/NPZ postprocessing outputs. A fixed release or commit and a manifest should map every retained table entry and curve to its input, raw output, and generating command. Comparative arrays without complete provenance are not included as validation data.

No proprietary data or restricted software was used. Quantum ESPRESSO [18] is freely available at `https://www.quantum-espresso.org`. Pseudopotentials were obtained from the SSSP library [20], freely available at `https://www.materialscloud.org/discover/sssp`.

Appendix A: Derivation of the Quantum Thermal Displacement Correlator

We derive Eq. (1) from the quantum harmonic lattice. In the second-quantized description, the displacement of atom i of mass M_i at lattice site $\mathbf{R}$ along Cartesian direction α is:

$$\hat{u}_i^\alpha(\mathbf{R}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{q},s} \sqrt{\frac{\hbar}{2M_i\omega_{\mathbf{q}s}}} e_i^\alpha(\mathbf{q}s)\big(\hat{a}_{\mathbf{q}s} + \hat{a}^\dagger_{-\mathbf{q}s}\big)e^{i\mathbf{q}\cdot\mathbf{R}}$$

where $\hat{a}^\dagger_{\mathbf{q}s}$ ($\hat{a}_{\mathbf{q}s}$) creates (annihilates) a phonon in branch s at wavevector $\mathbf{q}$, $e_i^\alpha(\mathbf{q}s)$ is the polarization vector, and N is the number of unit cells.

Computing the thermal average. The thermal correlator between atoms i (at the origin) and j (at $\mathbf{R}$) is $C_{ij}^{\alpha\beta}(\mathbf{R},T) = \langle \hat{u}_i^\alpha(\mathbf{0})\hat{u}_j^\beta(\mathbf{R})\rangle_T$ with $\langle\cdot\rangle_T = \mathrm{Tr}(e^{-\hat{H}/k_BT}\cdot)/Z$. Inserting the mode expansion:

$$C_{ab}^{\alpha\beta}(\mathbf{R},T) = \frac{\hbar}{2N_c\sqrt{M_aM_b}} \sum_{\mathbf{q}s}\sum_{\mathbf{q}'s'} \frac{e_a^\alpha(\mathbf{q}s)e_b^\beta(\mathbf{q}'s')}{\sqrt{\omega_{\mathbf{q}s}\omega_{\mathbf{q}'s'}}} \times \mathcal{A}(\mathbf{q}s,\mathbf{q}'s')e^{i\mathbf{q}'\cdot\mathbf{R}}.$$

where the bosonic correlator $\mathcal{A}$ collects the four terms from $(\hat{a}+\hat{a}^\dagger)(\hat{a}'+\hat{a}'^\dagger)$. Using the standard equilibrium averages:

$$\langle \hat{a}^\dagger_{\mathbf{q}s}\hat{a}_{\mathbf{q}'s'}\rangle_T = n_B(\omega_{\mathbf{q}s})\,\delta_{\mathbf{q}\mathbf{q}'}\delta_{ss'},$$
$$\langle \hat{a}_{\mathbf{q}s}\hat{a}^\dagger_{\mathbf{q}'s'}\rangle_T = \big(n_B(\omega_{\mathbf{q}s})+1\big)\delta_{\mathbf{q}\mathbf{q}'}\delta_{ss'}$$

where $n_B(\omega) = (e^{\hbar\omega/k_BT}-1)^{-1}$ is the Bose-Einstein occupation, and noting that anomalous averages $\langle\hat{a}\hat{a}\rangle_T = \langle\hat{a}^\dagger\hat{a}^\dagger\rangle_T = 0$ vanish, we obtain:

$$\mathcal{A} = (2n_B + 1)\,\delta_{\mathbf{q},-\mathbf{q}'}\delta_{ss'}.$$

The coth **identity.** We prove $2n_B(\omega) + 1 = \coth(\hbar\omega/2k_BT)$. Starting from the definition, with $x = \hbar\omega/k_BT$:

$$2n_B + 1 = \frac{2}{e^x - 1} + 1 = \frac{2 + e^x - 1}{e^x - 1} = \frac{e^x + 1}{e^x - 1}$$

Multiply numerator and denominator by $e^{-x/2}$:

$$\frac{e^{x/2} + e^{-x/2}}{e^{x/2} - e^{-x/2}} = \frac{\cosh(x/2)}{\sinh(x/2)} = \coth\left(\frac{\hbar\omega}{2k_BT}\right). \qquad \square$$

Using $\delta_{\mathbf{q},-\mathbf{q}'}$ to collapse the $\mathbf{q}'$ sum, using $e_i^{\alpha*}(\mathbf{q}s) = e_i^{\alpha}(-\mathbf{q}s)$ (time-reversal symmetry), and substituting, we arrive at the final result:

$$\boxed{C_{ab}^{\alpha\beta}(\mathbf{R},T) = \frac{\hbar}{2N_c\sqrt{M_aM_b}}\sum_{\mathbf{q},s}\frac{e_a^{\alpha*}(\mathbf{q}s)e_b^{\beta}(\mathbf{q}s)}{\omega_{\mathbf{q}s}} \times \coth\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right)e^{-i\mathbf{q}\cdot\mathbf{R}}.}$$

This is Eq. (1) with the same negative phase convention. At high temperature, $\coth(x/2) \to 2/x$ for $x = \hbar\omega/k_BT$, giving the classical mode-amplitude scaling k_BT/ω^2. At zero temperature, $\coth(x/2) \to 1$, leaving the zero-point contribution of every mode.

Appendix B: Distance Weighting and Correlation Range

The equal-time displacement correlator in Eq. (1) is a static canonical quantity. Its spatial behavior follows from the full low-frequency dispersion, eigenvectors, dimensionality, and boundary conditions. A finite dynamical phonon lifetime broadens a spectral function, but does not by itself imply a universal exponential bound on this equal-time correlator. In particular, acoustic modes can produce long-ranged or algebraic contributions. The previous relations $\tau = \hbar/(\gamma k_BT)$, $\xi_T = \bar{v}_s\tau$, and $K_{\min} \propto T^{-3}$ omitted the anharmonic matrix elements and phase space entering a three-phonon rate and are not used here.

The exponential $w(R;\xi_w)$ in Eq. (5) is retained only as a numerical weighting choice. It is not interpreted as a phonon mean free path or a proven thermal coherence length. The real-space IFC cutoff, supercell dimensions, and weighting length must instead be converged directly against branch-resolved frequencies. Material-specific $C_2(R,T)$ should also be plotted and tested against exponential and non-exponential forms before any decay law is claimed.

Appendix C: The SOTC Objective Function: Definition and Gradient

For the finite vector of supercell displacements, $\mathcal{Q}$ is a multivariable objective function rather than a functional. In the species-resolved shell notation of Eq. (2), it is

$$\mathcal{Q}(\mathbf{u}) = \sum_{a,b}\sum_{R<r_{\text{cut}}} w_{ab}(R;\xi_w)\Big[\bar{C}_{2,ab}^{\text{cell}}(R;\mathbf{u}) - \bar{C}_{2,ab}(R,T)\Big]^2,$$

$$w_{ab}(R;\xi_w) = \exp(-R/\xi_w).$$

The weight is a tunable numerical emphasis, not a formally optimal envelope derived from a phonon lifetime. Other positive weights are admissible, and ξ_w must be included in cutoff/sensitivity tests. The scalar shell objective also omits off-diagonal and anisotropic tensor components; extending the target to the full $C_{ab}^{\alpha\beta}(\mathbf{R},T)$ is possible but requires a correspondingly larger identifiability analysis.

Expansion in the phonon basis. The displacement field on the supercell of n atoms is expanded in phonon eigenvectors:

$$u_i^{\alpha} = \frac{1}{\sqrt{M_i}}\sum_{\mathbf{q}s}\left[Q_{\mathbf{q}s}\,e_i^{\alpha}(\mathbf{q}s)\,e^{i\mathbf{q}\cdot\mathbf{R}_i^0} + \text{c.c.}\right]$$

where $Q_{\mathbf{q}s}$ are complex mode amplitudes. The SOTC instantaneous correlator then reads:

$$\hat{C}_{ij}^{\alpha\beta}(\mathbf{R}) = \frac{1}{\sqrt{M_iM_j}}\sum_{\mathbf{q}s,\mathbf{q}'s'} Q_{\mathbf{q}s}Q_{\mathbf{q}'s'}^{*}\,e_i^{\alpha}(\mathbf{q}s)\,e_j^{\beta*}(\mathbf{q}'s')\,e^{i(\mathbf{q}-\mathbf{q}')\cdot\mathbf{R}_j^0}$$

Gradient of $\mathcal{Q}$. Differentiating with respect to the real mode amplitude $A_{\mathbf{q}s} = |Q_{\mathbf{q}s}|$:

$$\frac{\partial\mathcal{Q}}{\partial A_{\mathbf{q}s}} = 4\sum_{(i,j),\mathbf{R}} w(|\mathbf{R}|)\big[\hat{C}_{ij}^{\alpha\beta}(\mathbf{R}) - C_{ij}^{\alpha\beta}(\mathbf{R},T)\big]\frac{\partial\hat{C}_{ij}^{\alpha\beta}}{\partial A_{\mathbf{q}s}}$$

Initial amplitude. The harmonic mode amplitude provides an initialization:

$$A_{\mathbf{q}s}^{harm}(T) = \sqrt{\frac{\hbar}{2\omega_{\mathbf{q}s}}\coth\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right)}$$

Matching individual harmonic mode variances does not make one deterministic configuration reproduce the complete thermal covariance, nor does it prove that this point is stationary for the finite shell objective. Optimization is therefore still required even for a commensurate mesh. The final objective value must be reported together with initialization and optimizer-sensitivity tests; gradient descent from $A_{\mathbf{q}s}^{harm}$ is an algorithmic choice rather than an exactness result.

Appendix D: IFC Extraction by Ridge Regression: Full Derivation

Building the regression matrix. For each atom pair (i, j) with separation $\mathbf{R}$ within r_{cut}, Newton's third law gives $\mathbf{F}_i = -\sum_j \mathbf{\Phi}_{ij}(\mathbf{u}_j - \mathbf{u}_i)$ for collective SOTC configurations (where all atoms are simultaneously displaced). This relative-displacement form is algebraically equivalent to $-\sum_j \mathbf{\Phi}_{ij}\mathbf{u}_j$ when the acoustic sum rule $\sum_j \mathbf{\Phi}_{ij} = \mathbf{0}$ holds exactly, but is numerically more stable for collective displacements because it prevents spurious rigid-translation forces from contaminating the regression. Vectorizing over all n atoms and 3 directions yields:

$$\Delta\mathbf{F} = -\mathbf{G}\,\mathbf{p}.$$

Here $\Delta\mathbf{F} \in \mathbb{R}^{3nN_{\text{ens}}}$ stacks the reference-subtracted forces from all configurations, $\mathbf{p} \in \mathbb{R}^{N_{\text{IFC}}}$ contains the independent symmetry-reduced parameters, and $\mathbf{G} \in \mathbb{R}^{3nN_{\text{ens}} \times N_{\text{IFC}}}$. Before symmetry aggregation, a contribution associated with $p = \Phi_{ij}^{\alpha\beta}$ is proportional to the relative displacement $u_j^\beta - u_i^\beta$, not u_j^β alone.

Normal equations and ridge regularization. The least-squares solution minimizes $\|\mathbf{G}\mathbf{p} + \Delta\mathbf{F}\|^2$. When $N_{\text{IFC}} > 3n$ (underdetermined) or $\mathbf{G}^T\mathbf{G}$ is near-singular (correlated IFC parameters), Tikhonov regularization adds a penalty $\lambda\|\mathbf{p}\|^2$:

$$\begin{aligned}\mathbf{p}^* &= \arg\min_{\mathbf{p}} \left(\|\mathbf{G}\mathbf{p} + \Delta\mathbf{F}\|^2 + \lambda\|\mathbf{p}\|^2\right) \\ &= -\left(\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I}\right)^{-1}\mathbf{G}^T\Delta\mathbf{F}\end{aligned}$$

Rank and condition diagnostics. The singular values of the IFC design matrix are not, in general, identical to the normal-mode amplitudes: the matrix also contains bond geometry, parameter sharing, symmetry projection, and the stacking of configurations. No analytical condition-number advantage over a random/Gaussian control follows from exciting every phonon mode. Rank, singular values, and condition number must be computed from each actual symmetry-reduced matrix with a stated threshold. The effect of ridge regularization is reported through sensitivity to λ and error against a converged reference; it is not assumed to be unbiased.

Acoustic sum rule enforcement. The ASR, $\sum_j \Phi_{ij}^{\alpha\beta} = 0$ for each i, α, β, is imposed as a hard linear constraint via the Lagrange-multiplier method. Write the constrained optimization:

$$\min_{\mathbf{p}} \|\mathbf{G}\mathbf{p} + \Delta\mathbf{F}\|^2 + \lambda\|\mathbf{p}\|^2 \quad \text{subject to} \quad \mathbf{A}_{\text{ASR}}\mathbf{p} = \mathbf{0}$$

Solution via the KKT conditions gives the projected system: let $\mathbf{N}$ be the null-space basis of $\mathbf{A}_{\text{ASR}}$ (computed once by QR decomposition), write $\mathbf{p} = \mathbf{N}\tilde{\mathbf{p}}$, and solve the unconstrained problem for $\tilde{\mathbf{p}}$ using the reduced matrix $\tilde{\mathbf{G}} = \mathbf{G}\mathbf{N}$.

Central-force symmetry projection. For high-symmetry bonds (FCC nearest neighbors, rocksalt nearest neighbors), the IFC tensor can be decomposed into longitudinal and transverse components:

$$\Phi_{ij}^{\alpha\beta}(\hat{\mathbf{R}}) = A_{ij}\hat{R}_\alpha\hat{R}_\beta + B_{ij}\left(\delta_{\alpha\beta} - \hat{R}_\alpha\hat{R}_\beta\right)$$

where $\hat{\mathbf{R}} = \mathbf{R}/|\mathbf{R}|$ is the bond unit vector, A_{ij} is the longitudinal (bond-stretching) force constant, and B_{ij} is the transverse (bond-bending) force constant. This projection reduces the number of free IFC parameters per bond from 9 (general 3×3 tensor) to 2, improving the regression rank by a factor of $\sim$4.5 and dramatically stabilizing the fit.

Appendix E: Finite-Cell Truncation and IFC Identifiability

A finite SOTC cell and displacement cannot guarantee arbitrarily accurate phonons. Three independent errors must be controlled in practice.

First, a real-space cutoff truncates the IFCs. Its error is material dependent; long-range electrostatics in polar crystals require an explicit non-analytic term, and no universal electronic screening length establishes the cutoff used here. Second, the periodic cell must contain each retained interaction without uncontrolled image aliasing. Both errors require direct cell and cutoff convergence of the dispersion.

Third, the force inverse problem must be identifiable. With n atoms and N_{ens} configurations, the stacked geometry matrix has at most $3nN_{\text{ens}}$ independent rows. Space-group constraints, permutation symmetry, a central-force ansatz, and the acoustic sum rule reduce the IFC parameter count, while ridge regularization selects a stable solution when the data remain deficient. Exciting every normal mode does not alone guarantee that this parameter-space matrix has full column rank, and ridge regression does not add information.

Accordingly, each benchmark must report the raw and reduced parameter counts, matrix dimensions, singular values and threshold, numerical rank, condition number, and sensitivity to the physical priors. Accuracy must be assessed against converged reference IFCs and dispersions as the cell, cutoff, and number of configurations are increased. These empirical finite-data checks replace a universal theorem or high-temperature corollary.

Appendix F: Self-Consistency Map and Empirical Convergence

One outer iteration defines a numerical map $\mathcal{F} : C_2^{(n)} \mapsto C_2^{(n+1)}$: design configurations, evaluate forces,

fit constrained IFCs, construct phonons, and update the target correlators. Linear mixing forms

$$C_{2,\mathrm{mix}}^{(n+1)} = \beta\,\mathcal{F}(C_2^{(n)}) + (1-\beta)C_2^{(n)}.$$

The archived calculations stop when the norm of the change falls below 3×10^{-3} Å^2, subject to a material-specific minimum iteration count.

No contraction bound for $\mathcal{F}$ has been established. In particular, an estimate in terms of an effective Grüneisen parameter did not derive the derivatives of the displacement optimizer and regularized IFC fit, and cannot guarantee convergence. Numerical safeguards such as mixing, correlator floors, omission/rescaling of imaginary modes, and selection of a previous best iteration also modify the practical map. Accordingly, convergence of the scalar stopping metric must be distinguished from convergence to a unique physical IFC solution. Initialization, mixing, safeguard, and configuration-count sensitivity tests are required and are reported as empirical diagnostics rather than a fixed-point theorem.

Appendix G: Variational Scope of Deterministic Moment Matching

For a trial harmonic Hamiltonian H_0, the Gibbs–Bogoliubov inequality

$$F_{\mathrm{exact}} \le F_0 + \langle H - H_0\rangle_0$$

underlies variational self-consistent harmonic methods [9–11]. A method inherits this upper-bound property only if it evaluates the required expectation consistently and performs the appropriate variational minimization.

Matching a finite set of second moments,

$$\frac{1}{K}\sum_{k=1}^{K} u_i^{\alpha,(k)} u_j^{\beta,(k)} \simeq C_{ij}^{\alpha\beta},$$

does not by itself define a multidimensional Gauss–Hermite rule for general anharmonic forces or energies. It neither integrates all terms entering the Gibbs–Bogoliubov functional nor guarantees exponential quadrature convergence. The present SOTC algorithm also fits forces by ridge regression and stops on a correlator change; it does not minimize $F_0 + \langle H - H_0\rangle_0$. SOTC is therefore not assigned a rigorous variational upper bound, equivalence to a deterministic stochastic quadrature, or exponential convergence with configuration count.

The useful connection is limited to the use of harmonic covariance as organizing information for an effective harmonic model. Variational methods optimize a distribution, whereas SOTC selects finite configurations to match chosen moments and fits IFCs. Whether this deterministic moment design improves a force fit is an empirical question addressed by matched controls, not a consequence of the Gibbs–Bogoliubov inequality.

Appendix H: Harmonic Free Energy on the SOTC Cell

After self-consistency convergence, the renormalized frequencies $\omega^*_{\mathbf{q}s}(T)$ define the free energy of the best harmonic approximation. The single-mode partition function is:

$$\begin{aligned} Z_\omega &= \sum_{n=0}^{\infty} e^{-\hbar\omega(n+1/2)/k_BT} \\ &= e^{-\hbar\omega/2k_BT}\cdot\frac{1}{1-e^{-\hbar\omega/k_BT}} \\ &= \frac{1}{2\sinh(\hbar\omega/2k_BT)} \end{aligned}$$

using the geometric series. The single-mode free energy $F_\omega = -k_BT\ln Z_\omega = k_BT\ln(2\sinh(\hbar\omega/2k_BT))$. Summing over all $3N_{\mathrm{SC}}$ independent modes:

$$\boxed{F_{\mathrm{harm}}^{\mathrm{SOTC}}(T) = k_BT\sum_{\mathbf{q},s}\ln\!\left(2\sinh\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right)}$$

Thermodynamic quantities. Differentiating:

$$\begin{aligned} S_{\mathrm{harm}}(T) = k_B\sum_{\mathbf{q},s}\Bigg[&\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\coth\!\left(\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right) \\ &- \ln\!\left(2\sinh\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right)\Bigg]. \end{aligned}$$

$$C_V^{\mathrm{harm}}(T) = k_B\sum_{\mathbf{q},s}\left(\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right)^2\frac{1}{\sinh^2\!\left(\hbar\omega^*_{\mathbf{q}s}/2k_BT\right)}$$

$$\mathrm{ZPE} = F_{\mathrm{harm}}(0) = \sum_{\mathbf{q},s}\frac{\hbar\omega^*_{\mathbf{q}s}}{2}$$

For a monatomic cubic crystal, the one-component displacement convention used in the main text is

$$U_{\mathrm{iso}}(T) = \frac{\hbar}{6N_cM}\sum_{\mathbf{q},s}\frac{1}{\omega^*_{\mathbf{q}s}}\coth\!\left(\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right).$$

Compounds require the species-resolved eigenvector-weighted form.

Limiting behavior. At $T \to 0$: $2\sinh(x) \to e^x$, so $F \to \sum\hbar\omega/2$ (ZPE). At $k_BT \gg \hbar\omega$: $\sinh(x) \to x$, giving $F \to k_BT\sum\ln(\hbar\omega_{\mathbf{q}s}/k_BT)$ (classical Dulong-Petit). The heat capacity $C_V \to 3N_{\mathrm{SC}}k_B$ at high T, correctly recovering the Dulong-Petit law.

Appendix I: Higher-Order Anharmonic Terms: Scope

FC2 alone cannot supply anharmonic free-energy corrections or three-phonon lifetimes; those quantities require explicit higher-order calculations. Wick's theorem for a chosen Gaussian trial distribution is a standard identity, but it does not establish that the SOTC force fit minimizes the associated variational free energy or supply missing cubic and quartic force constants. Higher-order free energies, linewidths, lifetimes, and thermal conductivity are outside the validated scope of this manuscript.

Appendix J: Quasi-Harmonic Thermodynamic Identities and Convergence Requirements

Let $F(T,V) = E_0(V) + F_{\rm vib}(T,V)$ be the Helmholtz free energy. At external pressure p, the equilibrium volume minimizes the Gibbs free energy $F+pV$. At a stable minimum,

$$B_T = -V\left(\frac{\partial p}{\partial V}\right)_T = V\left(\frac{\partial^2 F}{\partial V^2}\right)_T > 0.$$

The volumetric expansion coefficient is $\alpha_V = V^{-1}(\partial V/\partial T)_p$; for a cubic solid the linear coefficient is $\alpha_L = \alpha_V/3$. Standard Maxwell relations then give [16]

$$C_P - C_V = TVB_T\alpha_V^2 = 9TVB_T\alpha_L^2.$$

These definitions correct the sign and factor-of-three/factor-of-nine ambiguities between common conventions.

For mode Grüneisen parameters

$$\gamma_{\mathbf{q}s} = -\frac{\partial \ln \omega_{\mathbf{q}s}}{\partial \ln V}, \qquad \gamma_{\rm eff}(T) = \frac{\sum_{\mathbf{q}s} c_{\mathbf{q}s}(T)\gamma_{\mathbf{q}s}}{\sum_{\mathbf{q}s} c_{\mathbf{q}s}(T)},$$

the cubic quasi-harmonic relation is

$$\alpha_V = \frac{\gamma_{\rm eff} C_V}{B_T V}, \qquad \alpha_L = \frac{\gamma_{\rm eff} C_V}{3B_T V},$$

with C_V and V expressed for the same amount of material.

No interpolation-error bound follows solely from a representative Grüneisen parameter and a 2% volume step. A third-order Birch–Murnaghan EOS requires enough independent static-energy points to determine and test its parameters; a two-volume IFC extrapolation cannot validate the phonon curvature. A converged protocol uses direct static energies and direct phonons at at least five volumes, tests alternative volume ranges and EOS forms, and reports leave-one-out residuals and propagated uncertainty in $V(T)$, $\alpha(T)$, $B_T(T)$, and $C_P(T)$. No subpercent quadratic-interpolation or IFC-extrapolation error is claimed without those tests.

Appendix K: Implementation and Reproducibility Details

The exact SSSP filenames used by the archived QE calculations were `Ag_ONCV_PBE-1.0`, `Au_ONCV_PBE-1.0`, `Mo_ONCV_PBE-1.0`, `Al.pbe-n-kjpaw_psl.1.0.0`, `Cu.paw.z_11.ld1.psl.v1.0.0-low`, `Mg.pbe-n-kjpaw_psl.0.3.0`, `O.pbe-n-kjpaw_psl.0.1`, `Pb.pbe-dn-kjpaw_psl.0.2.2`, `cl_pbe_v1.4.uspp.F`, `na_pbe_v1.5.uspp.F`, `Te_pbe_v1.uspp.F`, and `W_pbe_v1.2.uspp.F`. In QE input, the metallic smearing width was set with `degauss=0.02` Ry. In the Python interface, the archived central-force projection corresponds to `symmetrize_bonds=True`. The archived QHA class/keyword invocation is not evidence of thermodynamic convergence and is not used in the analysis.

The machine-readable result files are JSON and NPZ rather than uniformly plain text. A reproducibility archive should identify a single tagged commit and, for every number and plotted curve, provide the generating input, raw output, postprocessing command, and source-data file. External-method arrays without complete calculation-level provenance are not used for quantitative comparisons.

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